

Trace Metal cycling (E.g., Fe, Mn)

Dr. Sunil A. Patil
EES301

Introduction

- Iron (Fe) and manganese (Mn) are the most well-studied & prominent trace metal cycles.
- Generally considered to be **trace elements in open-water aquatic systems, but major elements in soils and sediments.**
- Although abundant (particularly Fe) in the Earth's crust, **they are often not readily available for organisms.**
- **Oxidized forms** such as iron oxides and hydroxides (e.g., $\text{Fe}(\text{OH})_3$, FeOOH , and Fe_2O_3), and manganese oxide (MnO_2) **are insoluble**, and settle in aquatic environments

- Fe redox cycling has been active and globally important throughout Earth's history (the massive deposition of iron-rich sediments, banded iron formations during the Archean Eon (>2.5 billion years ago) and parts of the Proterozoic Eon (2.5 to 0.54 billion years ago).
- At a 50-fold lower crustal abundance than Fe, Mn is the second most abundant redox-active metal
- Both serve as electron acceptors for microbes in anoxic environments (in particular, freshwater and marine sediments)
- Cycling involves redox transitions - oxidation and reduction reactions catalyzed mostly by prokaryotes - chemical or abiotic transformations are also prominent

Fe cycling

- 4th most abundant element in the Earth's crust
- **Ubiquitous element** found in the biosphere, lithosphere, and hydrosphere.
- A major component of cytochromes, hemoglobin, chlorophyll, other proteins - respiration, oxygen transport in the blood, photosynthesis, nitrogen fixation, and nitrate reduction.
- In its pure state, **it is a reactive metal that oxidizes readily in the presence of oxygen** (abiotic/spontaneous oxidation).
- **Fe interacts significantly with the cycles of other biologically important elements** such as carbon, sulfur, and phosphorus.

Fe availability

- Exists mostly in one of the two oxidation states: **reduced ferrous iron, Fe(II) or Fe^{2+}** **and oxidized ferric iron, Fe(III) or Fe^{3+}**
- Oxygen concentration, pH, and microbial activities affect their prevalence.
- Found in nature as solids in the form of **Fe(III)- and Fe(II)-bearing minerals** or as **ions dissolved in water**.

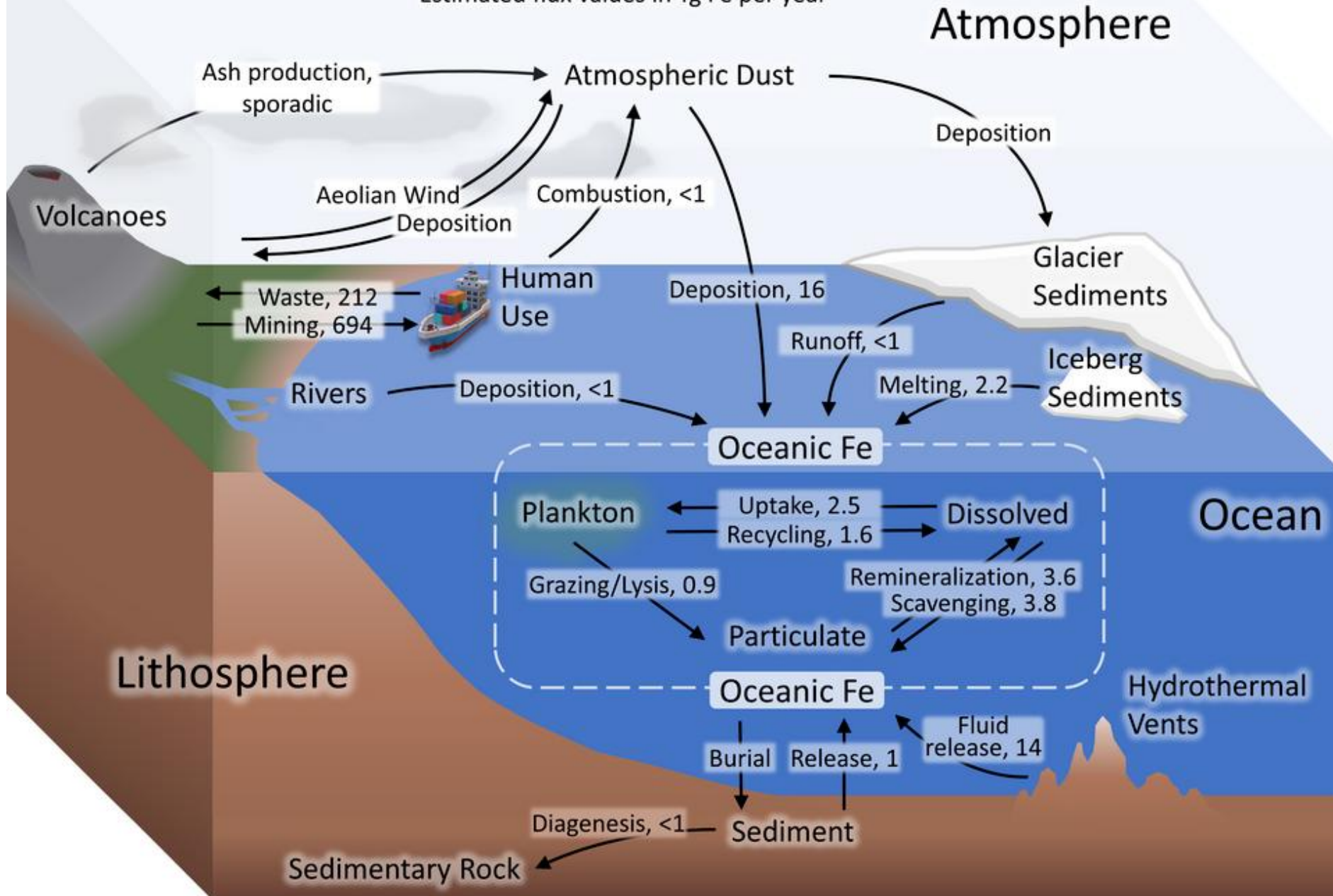


Most common:
Hematite (Fe_2O_3),
magnetite (Fe_3O_4),
wüstite (FeO), goethite
(FeOOH), siderite
(FeCO_3), Pyrite (FeS_2)

Iron oxides such as hematite (a and d) and goethite (b and e).
Iron sulfide such as Pyrite (c and f). © 2010 Nature Education

Global Iron Cycle

Estimated flux values in Tg Fe per year

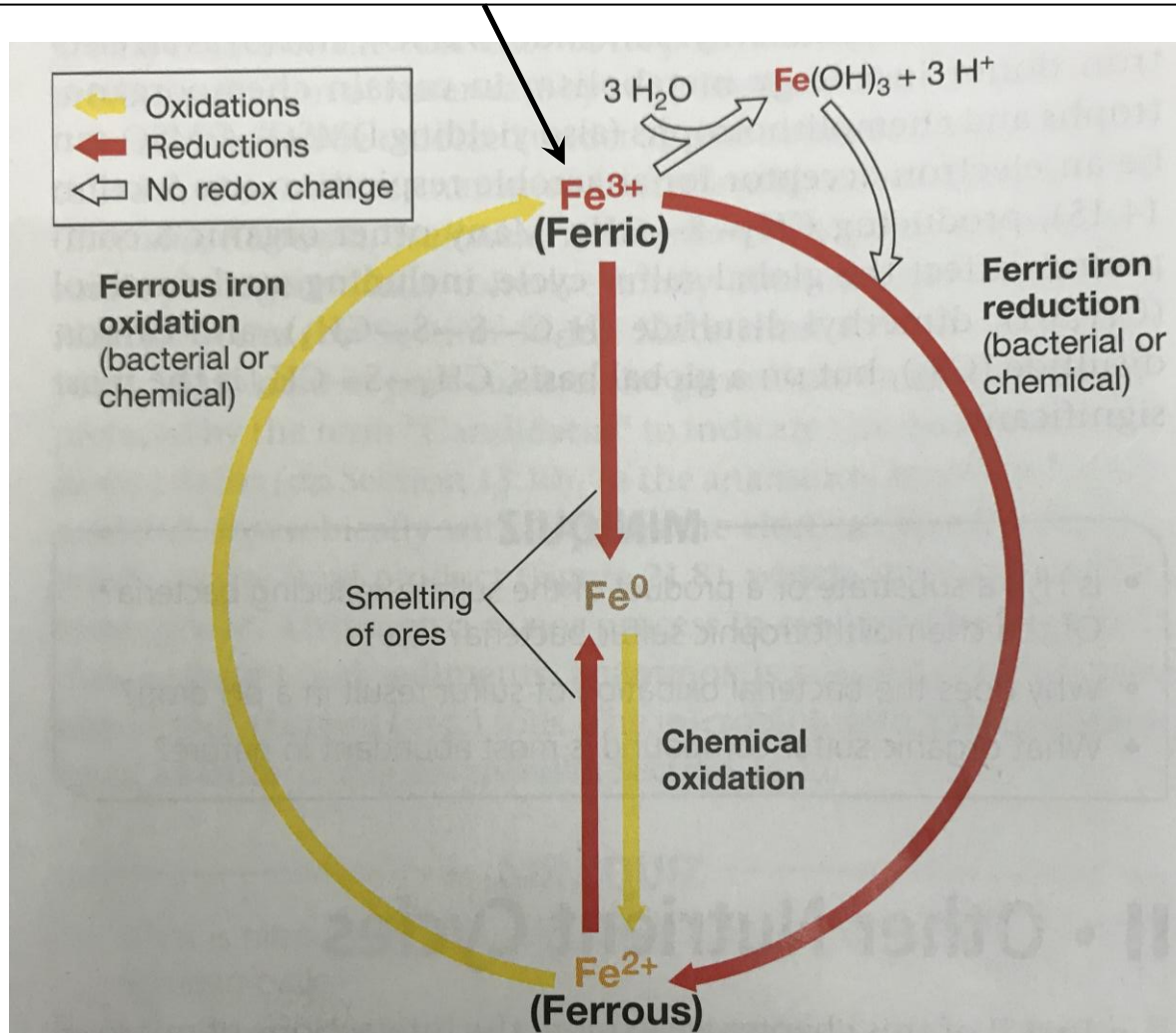


- Fluxes in Tg/year
- **Biosphere** – assimilation and transformations

https://en.wikipedia.org/wiki/Iron_cycle#/media/File:Iron_cycle7.png

Redox cycle of Fe

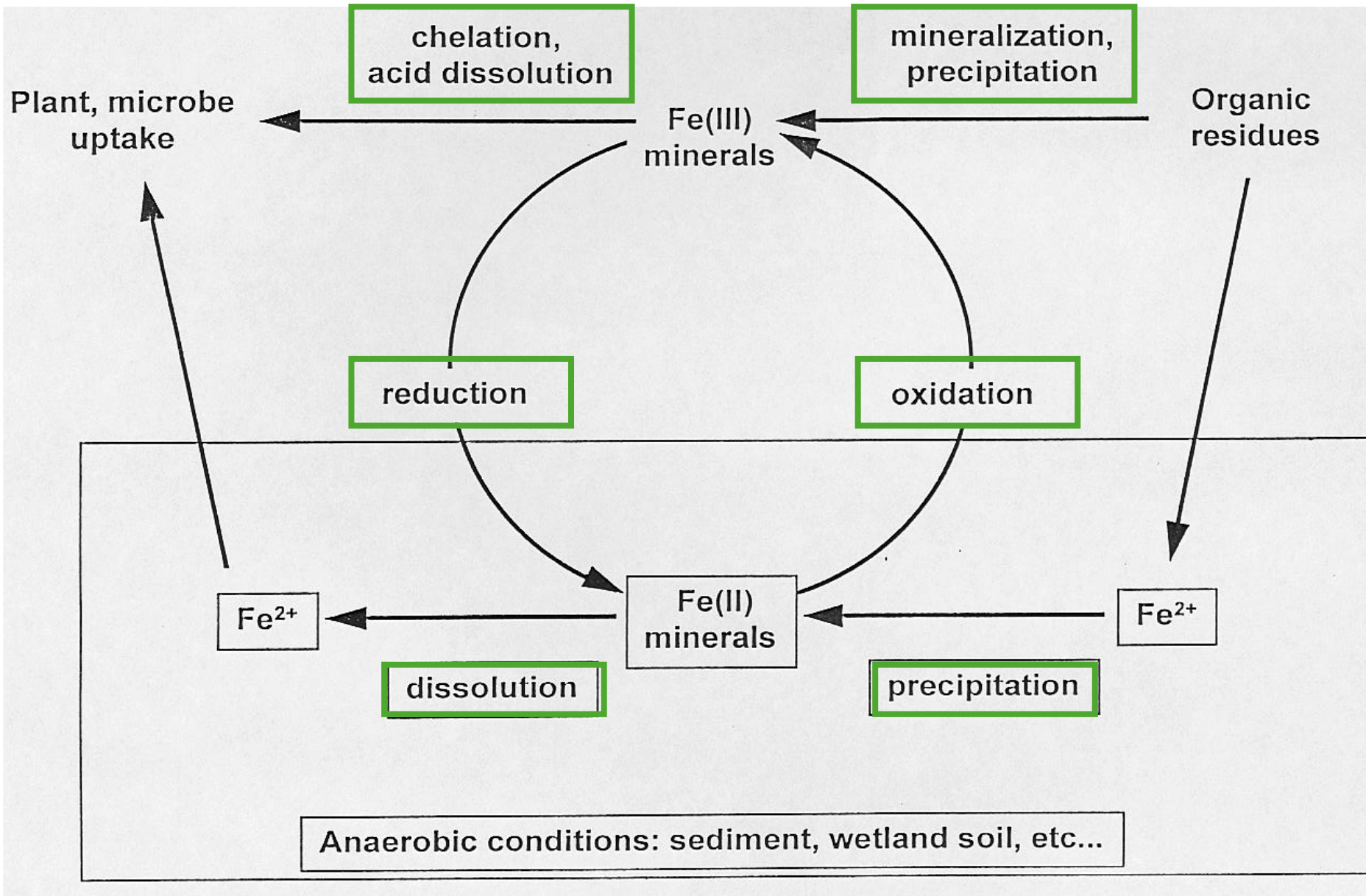
Weathering of minerals, volcanoes, iron dust, combustion sources, mining, industry wastewater, etc.



- One-electron oxidation & reduction reactions

**Earth's
ferrous wheel**

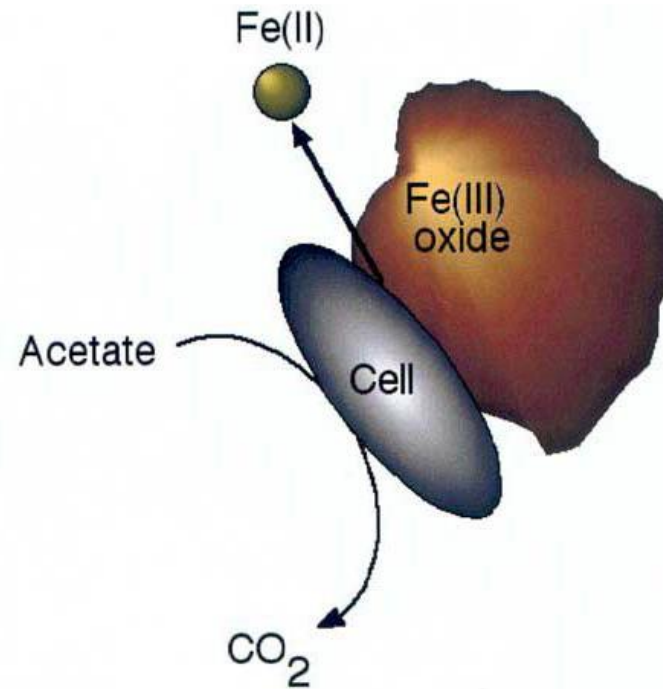
Microbial processes in iron cycling



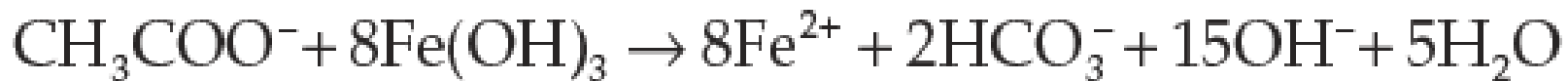
- Solubilization
- Assimilation
- Mineralization
- Immobilization
- Redox transformations

Fe³⁺ reduction

- Form of anaerobic respiration
- Fe³⁺ as an electron acceptor in the absence of other electron acceptors
- Fe³⁺ reduction coupled to oxidation of organics

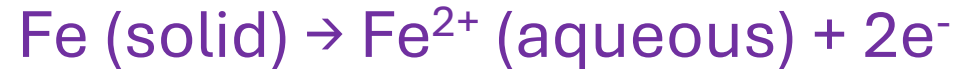


via extracellular electron transfer (direct/indirect)



Fe²⁺ oxidation under oxic conditions

- Abiotic oxidation - rapid at neutral pH (rusting); slower or negligible < 4 pH



- Ferrous iron (Fe²⁺) used as an electron donor by microbes - linked with oxygen reduction
- Fe²⁺ oxidation at low pH - microbial catalysis is manifold faster



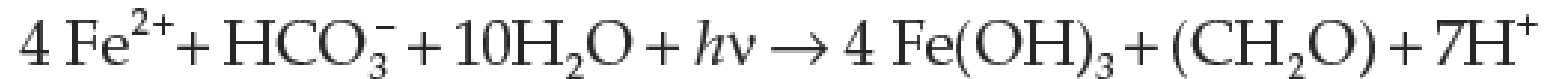
- Pyrite weathering/oxidation to form acid mine drainage



Fe²⁺ oxidation under anaerobic conditions

- **Phototrophic iron oxidation**

- Fe²⁺ as an electron donor in photosynthesis by some anoxygenic phototrophs (Purple non-sulfur bacteria)



- **Iron oxidation linked to nitrate reduction**

- Anaerobic respiration by some bacteria



Mn cycle

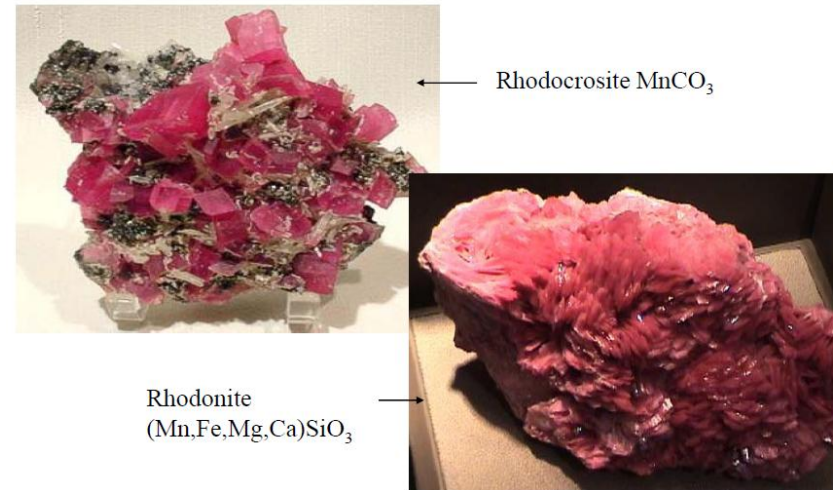
- Less abundant than Fe
- Mn is a cofactor for various enzymes
- Redox active states: **Mn(II)/(III)/(IV)**
- The Mn^{3+} is transient reactive redox species and quickly disappears, either by accepting or by donating an electron.
- **Major mineral forms (solids):** pyrolusite (MnO_2), braunite, psilomelane $(\text{Ba}, \text{H}_2\text{O})_2\text{Mn}_5\text{O}_{10}$ and others...



Pyrolusite



Braunite



Rhodocrosite MnCO_3

Rhodonite
 $(\text{Mn}, \text{Fe}, \text{Mg}, \text{Ca})\text{SiO}_3$

Mn reduction and oxidation



- Mn reducing bacteria in anoxic habitats



- Most microorganisms that reduce Fe(III) also can reduce Mn(IV)



- Many microbes are known to catalyze Mn oxidation under oxic conditions
- Energetically favorable for growth, but no organism has been conclusively shown to derive energy through this process

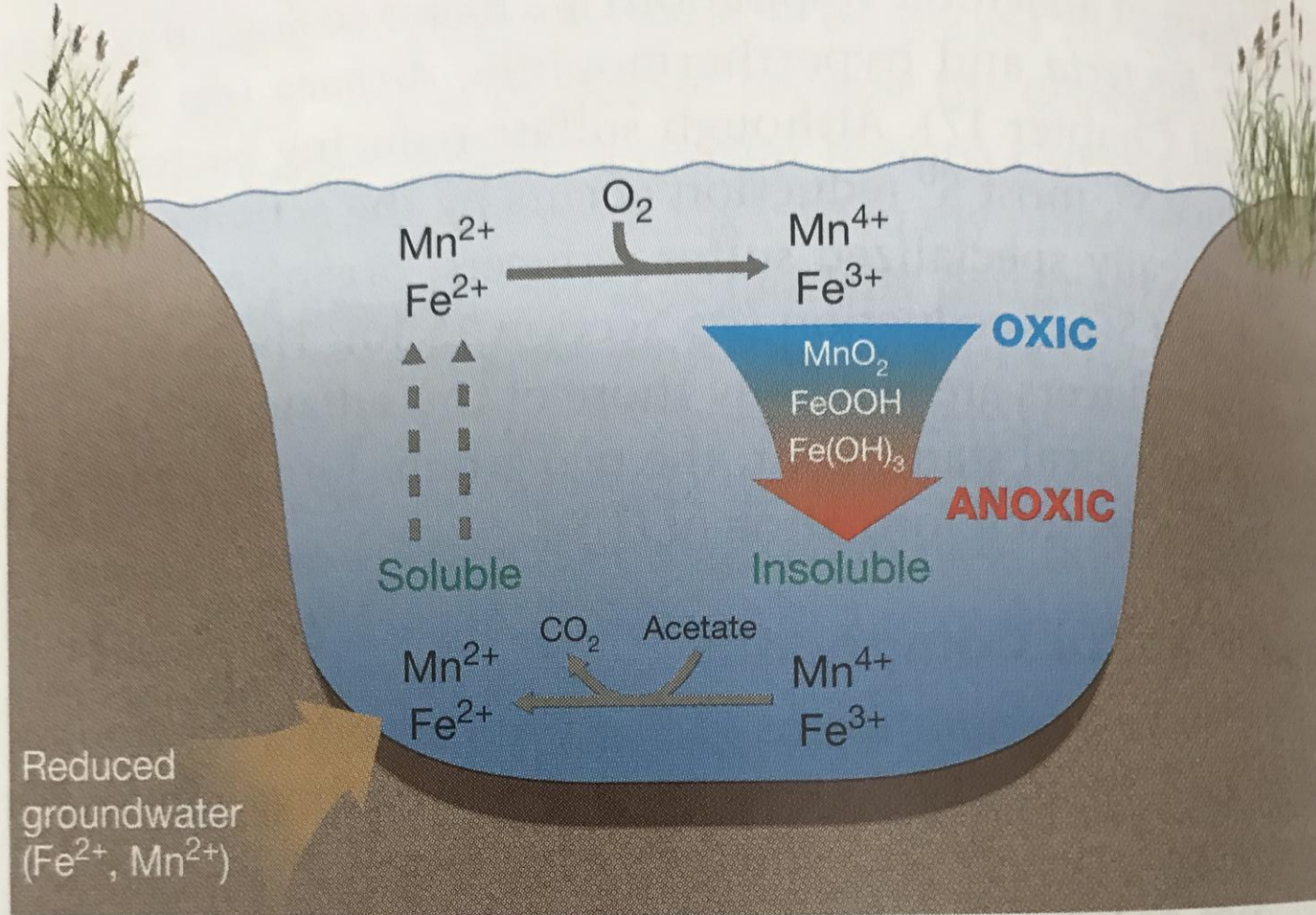


Figure 21.11 Iron and manganese redox cycling in a typical freshwater system. Iron and manganese oxides in sediments are used as electron acceptors by metal-reducing bacteria. The resulting reduced forms are soluble and diffuse into the oxic regions of the sediment or water column, where they are oxidized microbially or chemically. Precipitation of the insoluble oxidized metals then returns the metals to the sediments, completing the redox cycle.

Brock Biology of
Microorganisms, 15th edition

Significance of metal-reducing (& oxidizing) bacteria

- Soluble metal ions – bioavailability increases
- Biomineralization
- Fe(III)-reducers can outcompete sulfate-reducers and methanogens for electron donors and thus can limit the production of sulfides and methane in submerged soils, aquatic sediments, and the subsurface environments (anoxic environments)
- Biocorrosion
- Novel applications in wastewater treatment, bioremediation, energy production, bioelectronics, etc.